SOM. VIDYAVIHAR

Maximum Marks: 30	Semester: July 2024-November 2024 Examination: In-Semester Examination		Duration: 1.15 Hrs
Programme code: 01 Programme: L.Y.BTECH		Class: LY	Semester: VII (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP	
Course Code: 116U01E744	Name of the Course: Computer Simulation and modeling		

Question No.		Max. Marks	CO Mapped	BT Level
Q1	Consider the drive in restaurant where carhops take order and bring food to the car. Cars arriving according to the Inter-arrival distribution of cars. There are two carhops; C1 & C2. The distribution of their service time is also given in the table below. C1 is better in doing the job and works a little bit faster than C2. When the customer arrives, if he finds C1 free, the customer starts service immediately with C1. If C1 is not free but C2 is, then the customer starts service immediately with C2. If both are busy, customer starts service with the C1 to become free and if both are free then C1 gets the customer.	10	CO1 & CO2	AP & AN

Inter-arrival time (min)	2	3	4	5	6
Probability	0.18	0.25	0.27	0.17	0.13

C1 Service time (min)	2	3	4	5
Probability	0.17	0.24	0.29	0.30

C2 Service time (min)	3	4	5	6
Probability	0.18	0.22	0.30	0.30

Develop a simulation table and analyze the system by simulating the arrival and service of 10 customers. Assume that the first customer is arriving to system at 0^{th} time. Random digits for Inter-arrival time and service are given below.

Customer no	1	2	3	4	5	6	7	8	9	10
Random digit for Inter-arrival time	-	32	66	41	21	37	79	18	60	98
Random digit for service time	49	53	34	17	30	52	22	62	56	73

Q2	i) A medical examination is given in three stages by a physician. Each stage is exponentially distributed with a mean service time of 20 minutes. Find the probability that the exam will take 50 minutes or less. Also determine the expected length of the exam. ii) There are two workers competing for a job. Able claims an average service time that is faster than Baker's, but Baker claims to be more consistent, even if not as fast. The arrivals occur according to a Poisson process at the rate 2 per hour (l/30 per minute). Able's service statistics are an average service time of 24 minutes with a standard deviation of 20 minutes. Baker's service statistics are an average service time of 25 minutes, but a standard deviation of only 2 minutes. If the average length of the queue is the criterion for hiring, which worker should be hired? **OR** i) Draw flow chart of Execution of arrival & departure event for single channel queuing system using event scheduling. ii)Tabulate clock system state, future event list and cumulative statistics for single server queuing system using event scheduling approach for a given arrival and service details: Inter arrival time: 8, 6, 1, 8, 3, 8 Service time: 4, 1, 4, 3, 2, 4 Stop simulation when the clock reaches 30.	10	CO1 & CO2	UN & AP
Q3	**Answer Any Two:** i) Using K – S test, determine whether the hypothesis that following numbers are uniformly distributed on the interval [0, 1] can be rejected. Random numbers: 0.44, 0.81, 0.14, 0.05, 0.93. Use critical value: 0.565 for level of significance 0.05. ii) Test the following random numbers for independence by runs up and down test. Take $\alpha = 0.50$ and critical value $Z_{0.025} = 1.96$. Random numbers: {0.12, 0.01, 0.23, 0.28, 0.89, 0.31, 0.64, 0.28, 0.33, 0.93.	10	CO3	UN & AP

FY M.Tech

	iii) Simulation applications in Logistics and Transportation.			